

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 15

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark for each point.</i> • About 8 nights (1) • The running mean fluctuates widely with few samples but stabilises at 21-22 from 8 nights onwards; extra samples do not change it significantly (1)	2
01.2	<i>1 mark for each point.</i> • Moth activity varies with temperature, wind, rain and season (adults only fly for part of the year) so catches vary independently of population size (1) • Sampling at different conditions / times would make results not comparable (1) • Timing between samples should be chosen to detect the temporal variation of interest (eg weekly through the season) (1)	3
01.3	<i>1 mark for each point.</i> • Accuracy: how close a recorded result is to the real value (1) • Precision: the smallest interval between possible recorded values (eg a ruler marked in mm is more precise than one in cm) (1)	2
01.4	<i>1 mark for each point.</i> • A null hypothesis (eg no difference / no correlation) can be tested statistically and accepted or rejected at a stated confidence level (1) • Standardised techniques ensure results collected at different times, places or by different people are comparable (1) • Other variables are controlled or monitored so their effect can be assessed; allows the study to be repeated / checked (reliability) (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393; AQA spec 3.7.1.	

Question	Marking guidance	Marks
02.1	<i>1 mark for each point.</i> • Manufacture and use of CFCs and other ozone-depleting substances (ODSs) phased out then banned (1) • HCFCs to be phased out (by 2030); some essential uses (eg halon fire extinguishers in aircraft) still permitted (1) • A fund to help (developing) countries implement the Protocol; legally binding targets (unlike the 1985 Vienna Convention) (1)	3
02.2	<i>2 marks for each point (partial credit for a partly correct point).</i> • Aerosols: pump-action sprays / roll-on deodorants (alternative process), or hydrocarbon propellants (propane, butane) - flammable / fire risk (2) • Refrigerants: HCFCs (less stable so break down in the troposphere, but still some ozone depletion) then HFCs (no chlorine) - more expensive, less effective, powerful greenhouse gases (2)	4
02.3	<i>1 mark for each point.</i> • CFCs are drained from the appliance and incinerated / broken down at high temperature (1) • Products are CO₂ and acidic gases (HCl, HF) which are neutralised with crushed lime (1)	2
02.4	<i>1 mark.</i> • CFCs are persistent, so those already released continue to reach the stratosphere and destroy ozone for many years; some ODSs (eg dichloromethane) are not banned (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 112-113; AQA spec 3.2.1.3.	
03.1	<i>1 mark for each point.</i> • High mean wind velocities that are consistent in strength and direction (1) • A long fetch (distance of open water) over which waves build up, eg Atlantic coasts of the UK (1)	2
03.2	<i>1 mark for each point.</i> • Oscillating water column (eg Islay Limpet): waves force water up and down in a submerged chamber; the air forced in and out drives a turbine (1) • Or point absorber (floating buoy moving relative to a fixed base turns a generator); surface attenuator (Pelamis - hinged sections push pistons pumping fluid over a turbine); overtopping device (Wave Dragon - waves fill a reservoir above sea level which drains through turbines); oscillating wave surge converter (Oyster) (1)	2
03.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Equipment must withstand storms and corrosion (1) • Difficult to anchor off stormy coasts / in deep water (1) • Expensive to transmit electricity from isolated coasts to consumers (1) • Competition with established technologies (1)	2

Question	Marking guidance	Marks
03.4	<i>Up to 4 marks in total; award marks in proportion to the completeness of each point.</i> • Barrage: huge quantities of construction materials; reduces tidal range so intertidal mudflats used by wading birds are lost; changes sedimentation (erosion near turbines, deposition elsewhere), lower turbidity; pollutants concentrate in the lagoon; blocks migratory fish and shipping (up to 2) • Lagoon: similar but smaller scale; only part of the estuary enclosed so the most sensitive areas and main channel can be avoided (up to 2) • In-stream turbines: do not block the tidal flow so little impact on range or sediment; but noise affects cetaceans and many dispersed turbines could have a larger total impact; harness less energy (up to 2)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 210-211, 214-217; AQA spec 3.3.4.1.	
04.1	<i>Correct answer only.</i> • B - tiny pores between clay particles allow capillary rise; none in sand (1)	1
04.2	<i>1 mark for each point.</i> • Hygroscopic: held tightly by adhesive forces on particle surfaces; cannot drain and cannot be absorbed by plants (1) • Capillary: held in pores by surface tension; does not drain but can be absorbed by plants (1) • Gravitational: drains freely through large pores; available to plants only briefly as it passes (1)	3
04.3	<i>1 mark.</i> • Small round crumbs leave spaces giving good drainage, aeration and easy root penetration; large flat platy peds pack together reducing all three (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-169; AQA spec 3.2.5.	
05.1	<i>1 mark for each point.</i> • There is no gaseous / atmospheric reservoir of phosphorus (no equivalent of nitrogen fixation) (1) • Most phosphorus compounds have low solubility so little is available in soil water (1) • Phosphorus lost to marine sediments only returns through very slow processes: mountain building (uplift) and weathering of rocks (1)	3
05.2	<i>1 mark for each point.</i> • Fungal hyphae form a symbiotic relationship with plant roots and spread through a large volume of soil (1) • Greatly increasing the surface area for absorbing (poorly mobile) phosphate ions, passed to the plant in exchange for carbohydrates (1)	2
05.3	<i>1 mark for each point.</i> • Mining of calcium phosphate rock (and past mining of guano deposits, eg Peru, Nauru, now largely exhausted) (1) • Treated to make more soluble ammonium phosphate fertilisers, mobilising phosphorus into the environment (1) • Fertiliser runoff / eroded soil / sewage effluent carry phosphates into water bodies causing eutrophication (with nitrates); harvesting removes phosphorus from farmland (1)	3

Question	Marking guidance	Marks
05.4	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Use biological wastes (manure, sewage sludge, compost) as fertiliser to return phosphates rather than mining rock (1) • Breed crops that absorb phosphate more efficiently (1) • Maintain conditions for mycorrhizal fungi (less tillage, fewer fungicides, organic matter) (1) • Recover phosphate from sewage effluent by iron(III) sulfate precipitation; reduce soil erosion (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 156-157; AQA spec 3.2.4.4.	
06.1	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • $(68\,000 - 7\,000) / 68\,000 \times 100$ (1) • = 89.7% (accept 90%) (1)	2
06.2	<i>1 mark for each point.</i> • Inflow from the rivers was reduced (sometimes to zero) so less water entered while evaporation continued (1) • Evaporation removes pure water leaving the dissolved salts behind, so they became concentrated in a smaller volume (1)	2
06.3	<i>1 mark for each point.</i> • Salinity rose above the range of tolerance of the freshwater fish species / osmotic stress; pollution by pesticides, fertilisers and industrial waste concentrated in the lake (1) • Exposed lake bed of salt and pesticide-contaminated dust blown by wind onto farmland and settlements (health problems, soil salinisation) (1) • Loss of wetland habitats and wildlife; local climate more extreme (loss of the moderating effect of the water body) (1) • Loss of livelihoods; conflict between the countries sharing the rivers (1)	4
06.4	<i>1 mark for each point.</i> • Upstream abstraction for irrigation reduces the flow available downstream (1) • Upstream HEP dams hold water in summer for winter generation, reducing summer irrigation water downstream; growing populations increase demand (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 117-118; AQA spec 3.2.2.1-3.2.2.2.	
07.1	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Discharge of tank-washing water from tankers (1) • Oil rig / drilling accidents / leakage during drilling (1) • Pipeline leaks (1) • Leakage from storage tanks / refinery spills (1) • Waste lubricating oil from vehicles / industry washed into drains (1) • Other ship accidents (fuel oil) (1)	3
07.2	<i>1 mark for each point.</i> • Oil sticks feathers together so they lose their insulating properties and birds die of hypothermia (1) • Feathers lose buoyancy / waterproofing so heavily oiled birds drown (1) • Birds swallow toxic oil when preening; time spent cleaning feathers reduces time for feeding themselves / their young (1)	3

Question	Marking guidance	Marks
07.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • The surface film reduces the diffusion of oxygen from the atmosphere into the water, so aerobic organisms suffer from low dissolved oxygen (1) • Toxic components (benzene, xylene) dissolve and poison plankton near the surface (1) • Oil blocks light so algae / phytoplankton cannot photosynthesise (1) • Oil masks odours / narcotises so animals cannot find mates or food by smell (1) • Shore organisms (molluscs, crustaceans) are covered and asphyxiated (1)	2
07.4	<i>1 mark for each point.</i> • Dispersal / dilution by currents and waves is prevented so oil stays concentrated; in open water most spills disperse in days or weeks (1) • Inshore areas have more sensitive ecosystems (shores, reefs, mangroves, breeding sites) and more human activities (fishing, tourism) to be affected (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 258-259; AQA spec 3.4.3.2.7.	
08.1	<i>1 mark for the correct answer.</i> • $15 - (-18) = 33\text{ }^{\circ}\text{C}$ (1)	1
08.2	<i>1 mark for each point.</i> • Visible light passes easily through the atmosphere and is absorbed by the surface, which warms (1) • The warm surface emits (long-wave / far) infrared radiation (1) • Greenhouse gases (water vapour, CO_2, methane) absorb infrared better than the average for the atmosphere and convert it to heat (1) • The warm atmosphere emits infrared which is absorbed by the surface, and it reduces heat loss from land and oceans by conduction (1)	4
08.3	<i>1 mark.</i> • Any gas that is better at absorbing infrared radiation than the average for the atmosphere (1)	1
08.4	<i>1 mark for each point.</i> • Arriving: mainly ultraviolet, visible light and near infrared (short wavelength); leaving: mainly long-wavelength (far) infrared (1) • Gases that are transparent to the incoming short wavelengths can absorb the outgoing long wavelengths, so changing their concentration changes how much energy is retained (1)	2
08.5	<i>1 mark for each point.</i> • The energy and the processes it drives control climate, ocean currents and the hydrological cycle (1) • These determine the abiotic conditions (temperature, rainfall) that each species is adapted to, so a change alters where species can survive (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 87-89; AQA spec 3.2.1.1.	
09.1	<i>1 mark for each point.</i> • Natural processes may concentrate the material (eg adsorption onto sediments, bioaccumulation and food-chain concentration) (1) • It may be carried to locations where it is hazardous, such as fishing grounds or beaches used by people (1)	2

Question	Marking guidance	Marks
09.2	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • Physical state of the effluent (solution / solid) (1) • Ocean currents / river flow (1) • Effect of pH and oxygen on solubility (1) • Bioaccumulation and food-chain concentration (1) • Food sources and consumption by local people (eg shellfish, seaweed) (1) • Half-lives of the isotopes (1) • Meteorological conditions for any gaseous release (1) • Nature of the environment - geology, vegetation (1)	4
09.3	<i>1 mark for each point.</i> • Monitoring focuses on the sites most at risk of contamination (1) • No need to sample sites that were never likely to be polluted; sampling checks the accuracy of the predictions (1)	2
09.4	<i>1 mark for each point.</i> • It assesses the risk to the people most exposed because of their lifestyle before any health effects occur (1) • If their risk is acceptably low everyone else has a lower risk; if unacceptable, emissions are reduced to prevent future exposure (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-245, 303-304; AQA spec 3.4.3.1, 3.4.3.3.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • State of matter: solids deposit near the source, gases disperse widely in the atmosphere; density (lead dust, hydrogen cyanide at Bhopal) controls how far particles/gases travel • Persistence / half-life: persistent pollutants (organochlorines, CFCs, heavy metals, radioactive wastes with long half-lives) travel further, accumulate and need containment rather than dilution; non-persistent pollutants (sewage, organophosphates) can be allowed to degrade • Toxicity, specificity, mode of action (enzyme inhibition, neurotoxins) determine acceptable concentrations and which species are at risk; LD50 / MDAF / MDNF used to set doses • Solubility in water (nitrates disperse and leach - control by nitrate vulnerable zones) vs liposolubility (DDT, mercury bioaccumulate and biomagnify - control by bans / restrictions) • Reactivity: reactive pollutants may degrade quickly or form secondary pollutants (ozone, acid rain, PANs) - controlling primary pollutants (catalytic converters, FGD) prevents secondary pollutants • Adsorption onto clay / sediments immobilises metals and caesium; can be used deliberately (activated carbon filters, polymer adsorption) • Mutagenic / carcinogenic / teratogenic action means no safe dose; exposure kept As Low As Reasonably Achievable (ALARA) • Synergism (ozone + SO₂; smoke + SO₂) means pollutants must be considered together • Environmental features (temperature inversions, currents, pH, oxygen) interact with properties: Critical Pathway Analysis uses properties + environment to predict where pollutants go and to focus monitoring • Evaluation: properties of new chemicals can be predicted from similar ones (precautionary principle); understanding allows prevention rather than clean-up	9
10.2	<i>1 mark.</i> • The increase in the amount / concentration of a substance within an (individual) organism over time, because intake exceeds excretion or breakdown (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-240; AQA spec 3.4.1.	

Question	Marking guidance	Marks
11.1	To what extent can renewable energy resources replace fossil fuels in meeting the energy needs of a modern society? Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Fossil fuel features that shaped society: energy density, storage, transport, high temperatures, liquid vehicle fuels, established infrastructure • Renewable features: intermittent / unpredictable (wind, solar) vs predictable (tidal) vs controllable (HEP, biofuel, geothermal); low energy density; location; low pollution but manufacturing impacts • Technologies and developments for each resource with examples; new technologies improving usability • Storage and grid management: pumped storage, batteries, hydrogen economy, P2G, V2G, thermal storage, peak shaving; the role of electricity as a secondary fuel • Economics and politics: development costs, interest, subsidies, external costs of fossil fuels, falling renewable costs • Energy conservation as part of the solution; sustainability principles - natural systems run on low-density renewable energy • Judgement: electricity largely yes; heat, aviation, shipping and high-temperature industry harder; transition period with nuclear / CCS Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-235; AQA spec 3.3.2-3.3.4.	25

Question	Marking guidance	Marks
11.2	Soil is often described as a non-renewable resource. Discuss the causes and consequences of soil degradation and evaluate the strategies that can be used to manage soils sustainably. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Features of fertile soil: water, nutrients, air, organic matter, pH, biota, texture, structure, depth; slow rate of soil formation vs rapid erosion (dynamic equilibrium broken) • Causes of degradation: vegetation removal, ploughing vulnerable soils, overgrazing, reduced organic matter and biota, cultivating steep slopes, compaction, salinisation from irrigation, nutrient depletion, pollution • Types of erosion: rain splash, runoff, wind, slumping; consequences: reduced productivity, river and reservoir sedimentation, flooding, coastal sedimentation, particulates, desertification, landslides • USLE as a tool for estimating loss and testing management • Management: long-term crops, low / zero tillage, contour ploughing, tied ridging, terracing, stone rows, windbreaks, multicropping / strip cropping, mulching, organic matter, livestock control, drainage / pH / nutrient management, conservation of soil biota • Links to sustainable agriculture, carbon storage, hydrological cycle, circular economy • Evaluation and judgement on which strategies are most effective in different environments Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-169, 328-337; AQA spec 3.2.5, 3.5.1.3, 3.5.1.5.	25